To:
The Editor-in-Chief
Knowledge-Based Systems
Elsevier B.V.

Date: October 17, 2026

Subject: Submission of original research manuscript titled *"Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models"*

Dear Editor-in-Chief,

I am pleased to submit the above-titled original research paper for consideration for publication in *Knowledge-Based Systems*.

My background is in Civil Engineering, with specialization in computational mechanics, numerical methods, high-performance computing, and scientific computing. This perspective motivated the paper's integral formulation. In particular, I recognized a useful analogy with particle-based numerical methods such as smoothed particle hydrodynamics (SPH), where a field or operator is approximated by aggregating contributions from discrete particles through kernel-dependent weights. In the present work, the "particles" are abstract atoms and the weights are induced by a query-conditioned kernel and normalized into a read measure. This analogy helped me formulate attention, external retrieval, and expert routing in a common language of weighted discrete integration. The paper does not claim that neural attention or retrieval is an SPH discretization; rather, SPH provided an intuitive numerical-methods perspective for organizing the integral viewpoint and for separating the roles of the reference measure, kernel, normalization, and value map.

The manuscript develops a common valued-atom-space read abstraction for context attention, external retrieval, and parameter-expert routing. The formulation makes explicit the normalization and existence conditions for query-conditioned reads while allowing query-dependent expert outputs. It then derives a Lipschitz-margin condition for post-hoc insertion and exact softmax-tail certificates with an error-budgeted truncation rule; both are shown to be substantially looser than practical behavior in controlled benchmarks. The paper also establishes exact anti-atom cancellation and quantifies its sensitivity to key perturbations. CounterFact-500 experiments report held-out ROC calibration of a retrieval activation threshold, while matched GPT-2 experiments place the external read beside a rank-one parametric reference and a GRACE reimplementation. The paper presents these results as empirical and mathematical calibrations rather than as a new editing method or a claim of superiority.

The experiments are intentionally presented as mechanism studies. The work includes five-seed synthetic insertion and tri-space routing studies, CounterFact retrieval and calibration, signed-read stress tests, and first-token GPT-2 steering diagnostics. The manuscript explicitly states its limitations: the synthetic and routing tasks are templated, the truncation certificate requires dense scoring, the steering intervention is deliberately conditional on retrieval, the GRACE and rank-one comparisons use a first-subtoken scorer, and the paper does not provide certified parametric unlearning or a comprehensive comparison with methods such as SERAC, MEMIT, MEND, or WISE. The complete codebase, benchmark definitions, result files, and reproduction scripts are publicly accessible at `https://github.com/mossaiby/RoAS`.

I confirm that the manuscript is original and is not under consideration elsewhere, and that the author declares no competing financial or personal interests. During manuscript preparation, I used generative AI as an editorial and technical-writing aid to help make the presentation more complete and to refine terminology used in machine learning, numerical analysis, and computational science. The research idea, numerical-

methods analogy, mathematical framing, experiments, interpretation of results, and final responsibility for the manuscript are my own; all AI-assisted text was reviewed, revised, and validated by me. This use is disclosed in the manuscript in accordance with Elsevier policy.

The paper may be relevant to readers interested in neural memory, retrieval, sparse routing, and mathematical descriptions of attention. Thank you for considering the manuscript.

Sincerely,

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GitHub Repository: `https://github.com/mossaiby/RoAS`